

MULTI-EXPOSURE FUSION WITH CNN FEATURES

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ABSTRACT

Multi-exposure fusion (MEF) is a widely used approach to high dynamic range imaging. The selection of features for fusion weight calculation is important to the performance of MEF. In this paper, we investigate the effectiveness of convolutional neural network (CNN) features for MEF. Considering the fact that there are no ground-truth images in MEF to train an end-to-end CNN, we adopt the pre-trained networks in other tasks to extract the feature. Both the selection of network and the selection of convolution layer are studied. With the extracted CNN feature map, we compute the local visibility and consistency maps to determine the weight map for MEF. The proposed method works well for both static and dynamic scenes. It exhibits competitive quantitative measures, and presents perceptually pleasing MEF outputs with little halo effects.

Index Terms— Multi-exposure fusion, High dynamic range imaging, CNN feature

1. INTRODUCTION

Natural scenes usually span a high dynamic range (HDR), which is challenging for digital single-lens reflex cameras to capture. High dynamic range (HDR) imaging aims to extend the dynamic range of digital cameras by capturing an image sequence with multiple exposures. There are two categories of methods to generate an HDR-like image: multi-exposure fusion (MEF) [1–3] in image domain, and HDR content reconstruction via camera response function (CRF) estimation and tone mapping [4,5]. Due to the complexity of recovering CRF and designing tone mapper [5], MEF is more preferably used in most consumer grade devices such as camera phones and digital cameras for its simpler implementation.

Many works on MEF have been reported in past decades. In [6], the source images are segmented into non-overlapped patches and fused based on the rule of max-entropy. The blocking artifact is avoided by blending neighboring patches. In [2], an image patch is decomposed into three components: strength, structure and intensity, and the fusion is performed on each component. This scheme is extended [2, 3] to fuse dynamic scenes via calculating the structural similarity between corresponding spatial patches. The MEF method [1]

computes the weight map via three image quality measures (contrast, color saturation and well-exposedness) and fuses the images in an efficient multi-resolution framework. However, this method only works for static scenes. Zhang *et al.* [7] proposed a gradient based MEF method for both static and dynamic scenes. The gradient magnitude is used for setting the fusion weight and the gradient direction is used to detect the moving objects across the sequence. Similarly, Gu *et al.* [8] utilized the structure tensor in gradient domain to compute the weight. In [9], Liu *et al.* proposed a SIFT descriptor based MEF method, which exhibits superior performance to [7] due to its more effective feature representation. In single-scale fashion, the edge-preserving filters [10–13] are used for smoothing weight maps to alleviate artifacts.

Feature extraction plays a pivotal role in determining the weight map for MEF. Most MEF methods [1, 7, 8] implicitly or explicitly incorporate the feature extraction in the design of fusion weights. For example, the second order Laplacian filter, one-order gradient operator and SIFT descriptor are used as the feature extractors in [1, 7, 9], respectively. All the features used in current MEF methods are hand-crafted features. In this paper, we exploit the possibility of utilizing convolutional neural network (CNN) [14–16] features for MEF, which is the first of its kind to the best of our knowledge. We found that shallow-layer CNN features can be leveraged for both static and dynamic MEF, achieving state-of-the-art MEF performance.

2. RELATED WORK

CNN has achieved state-of-the-art results in high-level vision problems [17], such as classification, segmentation and object detection, owing to its powerful discriminative feature learning ability. Recently, CNN has also been successfully used in many low-level vision problems such as super-resolution, denoising, and enhancement [15, 16, 18, 19], leading to impressive outcomes. Guided by the labeled training samples, deep CNNs can extract more effective features than conventional hand-crafted features.

However, few works have been reported to apply CNN for MEF. This is mainly because of the lack of ground-truth output for a given multiple exposure sequence so that the end-to-end learning of CNN cannot be adopted. Recently, Prabhakar

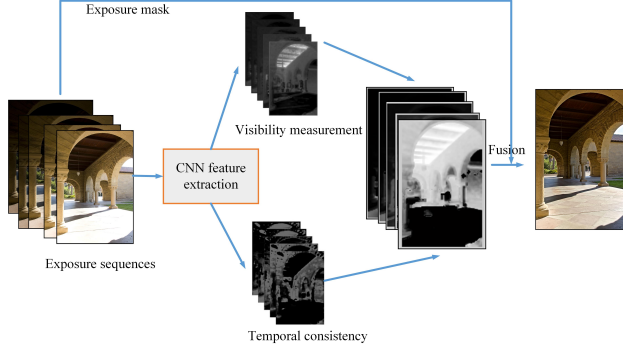


Fig. 1. Flowchart of the proposed CNN feature based multi-exposure fusion method.

et al. [20] proposed two ways for CNN based MEF. One is to select the best result produced by several MEF methods as the “ground-truth” via a subjective study. The other is to learn the CNN by optimizing a no-reference image quality metric defined in [21]. Although the authors claimed an un-supervised learning via the quality metric [22], they actually used the optimized result in [22] as “ground-truth” for supervision. Besides, this method is not applicable to dynamic scenes. In our work, we do not train a CNN to generate the MEF output; instead, we exploit the features of pre-trained CNNs to compute weight maps for simple yet effective MEF, which works for both static and dynamic scenes.

3. PROPOSED METHOD

This section presents the proposed CNN feature based MEF method. The flowchart of our method is shown in Fig. 1, which contains four major components: CNN feature extraction, pixel visibility measurement, temporal consistency check, and exposure mask calculation.

3.1. Feature extraction and visibility measurement

In general, there are two types of deep CNNs available for feature extraction: regression network (usually for low-level vision problems) and classification network (usually for low-level vision problems). In our application, dense features are required because we need to compute a weight at each pixel. For regression networks such as the denoising network [15] and super-resolution network [16], usually dense features can be obtained at each layer. For classification networks such as the VGG-Net [14], the dense features can only be obtained at shallow layers, which are however guided by the deeper layer sparse features with certain high level semantic information (e.g., discriminative parts, class labels). The selection of network and layer will be discussed in Section 4.1.

Let $I_i, i = 1, 2, 3, \dots, K$, be a sequence of K multi-exposure images. The feature map of each source image is

extracted as

$$\mathbf{F}_i(x, y) = \text{CNN}(I_i)(x, y), \quad (1)$$

where $\text{CNN}(\cdot)$ is a pre-trained deep network at some layer. For one pixel (x, y) , we can obtain a feature vector whose dimension is the number of filters used in that convolutional layer. The response of a pixel to CNN filters can indicate whether this pixel is important or informative for image representation. Therefore, the strength of feature vector $V_i(x, y)$ can be used to determine the visibility of pixel $I_i(x, y)$. We measure the visibility of pixel $I_i(x, y)$ as the L_1 norm of $V_i(x, y)$:

$$V_i(x, y) = \|\mathbf{F}_i(x, y)\|_1. \quad (2)$$

Compared with L_2 norm, the L_1 norm is selected due to its simpler calculation. The degree of visibility will affect value of weight to be assigned to this pixel. The pixels that have good local contrast are usually given bigger weights.

3.2. Temporal consistency

When the source images are taken in dynamic scenes, motion detection is needed to avoid the ghosting effect. In our framework, the motion detection can be easily implemented via calculating the Euclidean distance of two normalized feature vectors. Denote by $\bar{\mathbf{F}} = \frac{\mathbf{F}}{\|\mathbf{F}\|}$ the L_2 normalized feature of $\mathbf{F}(x, y)$ to remove the impact of exposure difference. The distance of two feature vectors at a pixel of two images is computed as:

$$s_{ij}(x, y)^2 = \|\bar{\mathbf{F}}_i(x, y) - \bar{\mathbf{F}}_j(x, y)\|_2^2, \quad (3)$$

$s(x, y)^2$ measures the similarity of two vectors. A smaller $s(x, y)^2$ represents stronger temporal consistency. When there is motion, the consistency will be destroyed, resulting in a smaller similarity value. We use a Gaussian kernel to map the similarity between $\bar{\mathbf{F}}_i(\mathbf{x}, \mathbf{y})$ and $\bar{\mathbf{F}}_j(\mathbf{x}, \mathbf{y})$ into the range of $[0, 1]$:

$$S_i(x, y) = \sum_{j=1}^K \exp \frac{-s_{ij}(x, y)^2}{2\sigma^2}, \quad (4)$$

where σ is a constant and is set as 0.05 in this paper. A larger weight should be given to the pixels which are temporally consistent.

3.3. Fusion

With the visibility and similarity weight maps V and S , we can get the final weight map $W_i(x, y)$ as follows:

$$W_i(x, y) = \frac{V_i(x, y) \times S_i(x, y) \times M_i(x, y)}{\sum_{j=1}^K V_i(x, y) \times S_i(x, y) \times M_i(x, y) + \alpha}, \quad (5)$$

where $M_i(x, y)$ is the exposure mask which is computed on the pixel intensity. To avoid division by zero, we add a small

Table 1. The MEF-SSIM scores by different networks at different layers on static scene dataset [2]

Network Type	Denoising [15]				Super-resolution [16]				Classification [14]			
Feature Layer	1	3	10	18	1	3	10	18	1	3	10	18
MEF-SSIM	0.869	0.970	0.969	0.965	0.867	0.957	0.846	0.930	0.969	0.620	0.610	0.560

coefficient α with value 10^{-10} . The widely used mask is the hat function defined as follow:

$$M_i(x, y) = \begin{cases} 1, & \beta < I_i(x, y) < 1 - \beta, \\ 0, & \text{else}, \end{cases} \quad (6)$$

where $\beta \in [0, 1]$ is a parameter controlling the exposure quality when the input images are normalized. It can effectively remove the poor exposure pixels. We choose β as 0.2 in our implementation. In practice, the weight map W can be smoothed by the edge-aware recursive filter [12] to further reduce the halo effect. Finally, we fuse the images as follows to produce the MEF output I_f :

$$I_f(x, y) = \sum_{i=1}^K I_i(x, y) \times W_i(x, y). \quad (7)$$

4. EXPERIMENTAL RESULTS

In Section 4.1, we discuss the selection of CNN networks and the associated layers for feature extraction. Then we compare the performance of CNN features with traditional hand-crafted features in Section 4.2. Finally, we compare our method with state-of-the-art MEF methods in Section 4.2. In all our experiments, we adopt the metric MEF-SSIM proposed in [21] as a quantitative measure to evaluate the performance of MEF methods on static scenes.

4.1. The selection of network and layer

We adopt two regression networks, denoising network DnCNN [15] and super-resolution network VDSR [16], and one classification network VGG19 [14], in the experiments. By using the features at different layers to compute the weight map W , the MEF image can be computed and the MEF-SSIM scores of different networks on the static scene dataset [2] are listed in Table 1. Note that for the VGG19 network, the features at deeper layers become sparser due to pooling, and we interpolate the feature maps to calculate the weight for each pixel. We can have the following observations from Table 1.

Table 2. The MEF-SSIM scores of CNN and traditional features on the static scene dataset [2]

Feature type	CNN	SIFT	Gabor
MEF-SSIM	0.969	0.952	0.900

First, the regression networks DnCNN and VDSR achieve

their best MEF-SSIM indices at shallow layers (more specifically layer 3) but not the first layer. The deeper layers become more task specific for denoising and super-resolution, but cannot bring benefit for MEF. Second, the classification network VGG19 achieves its best MEF-SSIM index at the first layer. With the increase of layers, the performance decreases rapidly. This is because the feature maps are getting sparser and sparser due to the spatial pooling in VGG19 network so that the weight map becomes less accurate. On the other hand, though the layer 1 features are very shallow features, they are guided by the deeper high level semantic features in training, which can still capture information of image structures. This is one advantage of classification networks over regression networks. Third, layer 3 DnCNN features, layer 3 VDSR features and layer 1 VGG19 features achieve very similar MEF-SSIM indices (the visual quality of their fused images is also similar). Considering that layer 1 features need much less computational cost and storage space, we select layer 1 VGG19 features as our feature extractor.

4.2. Comparison between CNN and traditional features

We then compare the effectiveness of layer 1 VGG19 features with traditional Gabor features and SIFT features. The objective MEF-SSIM indices by the three types of features are shown in Table 2. It can be seen that CNN features bring much better MEF performance than the hand-crafted features. Due to the limit of space, we do not show the visual comparison here, while CNN features indeed bring better perceptual quality of MEF images.

4.3. Comparison with state-of-the-art methods

We then compare our method with state-of-the-art MEF algorithms. On static sequences, we compare it with “Ma” [3], “Mertens” [1], “Gu” [8], “Shutao” [11], “Shen” [24], “Raman” [10] and “Li” [23]. Since some methods cannot be applied to dynamic sequences with motion, we compare our method with “Gallo” [25], “Li” [26], “Ma” [3], “Photomatix”, and “Sen” [27] on dynamic data. The codes of competing methods “Ma” [3], “Mertens” [1] and “Sen” [27] are from the original authors and we use their default settings. Other MEF results are copied from [2] or from the original papers. The “Photomatix” is commercial software from the website¹. The MEF-SSIM indices by competing methods on the static scene dataset [2] are listed in Table 3. Fig. 2 and Fig. 3 compare the

¹<https://www.hdrsoft.com//>

Table 3. The average MEF-SSIM scores of different methods on the static scene dataset [2]

Methods	[8]	[23]	[11]	[10]	[12]	[24]	[1]	[3]	Ours
MEF-SSIM	0.910	0.944	0.965	0.852	0.960	0.753	0.975	0.977	0.969

**Fig. 2.** The MEF results by competing methods on a static scene. From (a) to (f): results by “Ma” [3], “Mertens” [1], “Gu” [8], “Shutao” [11], “Shen” [24], and “Ours”.

MEF results by representative methods on a static scene and a dynamic scene, respectively. Please note that by far there is not a reliable objective quality measure for MEF results on dynamic scenes yet.

From Table 3, one can see that our method produces very competitive MEF-SSIM measures. Its average MEF-SSIM index is only lower than methods [1] and [3]. It should be noted that how to design a faithful objective quality measure for MEF is still a challenging issue. We found that some images with high MEF-SSIM scores exhibit obvious displeasing artifacts. The average run-time of our method is 0.90s, comparable with [1] (0.87s), but much faster than [3] (2.33s) on a computer with 4G Hz CPU and 32G RAM. The run-time of other methods is not available due to the lack of source codes.

Fig. 2 compares the MEF results on a static scene. The methods “Ma” [3], “Mertens” [1] suffer from the detail loss in bright regions, as well as the low contrast in dark regions. Method “Shen” [24] and “Gu” [8] shows obvious over-enhancement. Though method “Shutao” [11] shows overall good visual quality on this image, it exhibits slightly lower contrast compared with our method. For the dynamic scene in Fig. 3, method “Gallo” [25] suffers from obvious color artifacts; methods “Li” [26] and “Ma” [3] lose some details in the roof of the arch. Method “Photomatix” exhibits severe over-exposure problem. The result by the tone mapping method

**Fig. 3.** The MEF results by competing methods on a dynamic scene. From (a) to (f): results by “Gallo” [25], “Li” [26], “Ma” [3], “Photomatix”, “Sen” [27], and “Ours”.

“Sen” [27] is not natural with over-artistic effect. Our method provides a good balance between deghosting and detail preserving, proving the effectiveness of CNN feature on motion detection in multi-exposure sequence.

5. CONCLUSIONS AND FUTURE WORK

This paper made the first attempt to exploit the CNN features for weight design in MEF. Pre-trained CNN networks on other tasks were used to extract the features of each image. With these CNN features, the visibility and temporal consistency of each pixel in each image were defined, based on which the weights can be computed for MEF. We investigated the performance of regression networks and classification networks, and found that the very shallow layers of classification network can lead to desirable MEF outputs with low cost. Overall, the proposed method is simple and efficient to implement, and presents competitive results with state-of-the-arts on both static and dynamic scenes. In the future work, we will investigate how to fine-tune the CNN features to make them more effective for MEF applications.

6. ACKNOWLEDGEMENT

This work is supported by China NSFC grant (no. 61672446).

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